

Loan Gateway — Business Requirements Document (Lean)

Document version: 1.0 **Date:** May 2026 **Prepared by:** [Customer] **For:** [Web development vendor]
Status: Issued for fixed-price quotation

1. Project Summary & Canonical References

[Customer] is engaging [Vendor] to build a production-ready online loan application gateway for a Bangladesh-based Non-Banking Financial Institution. A complete working prototype already exists, and **the prototype is the canonical visual and behavioural specification of this engagement.**

All visual design, interaction patterns, microcopy, colour palette, typography, bilingual translations (English + বাংলা), modal flows, calculator logic, eligibility computation, multi-stage forms, validation rules, and tracking-number format are finalised in the prototype. **No design phase, wireframing, or UX-discovery work is required.** [Vendor] is asked to engineer a production version aligned with the prototype.

Reference	Location
Live prototype	https://rahilearns.github.io/Sample-Online-Platform/
Source code repository	https://github.com/Rahilearns/Sample-Online-Platform
Branding placeholder	The prototype uses "IDLC" as a placeholder brand. Production copy and branding will be supplied by [Customer].
Sample content disclosure	Every page in the prototype carries a "Sample content for prototype" note on the policy sections. These are placeholders. Production policy text will be supplied by [Customer] before launch.

The vendor may review, fork, and adapt any portion of the front-end code (HTML/CSS/JavaScript) at no licence cost.

2. Scope

2.1 In scope — Phase 1 MVP (fixed price)

Pages (11 total, no additional pages):

1. `index.html` — Marketing home (nav, hero, stats, news, why-us, quick links, footer; floating Customer Login + Apply for Loan buttons)
2. `apply.html` — Choose Individual / Business
3. `apply-individual.html` — Choose Home / Car / Personal Loan with modal pre-filter
4. `apply-business.html` — SME (with modal) / Corporate (external link)
5. `apply-home-loan.html` — Home Loan product page + OTP-gated application form + EMI calculator
6. `apply-car-loan.html` — same shape, Car Loan content
7. `apply-personal-loan.html` — same shape, Personal Loan content
8. `apply-sme-loan.html` — same shape, SME Loan content
9. Admin dashboard — list submitted applications, search by tracking number, view detail, export CSV
10. Admin login page
11. 404 / generic error page

Stubbed backend integrations for Phase 1:

- **OTP:** hardcoded value `1234` (no real SMS sent)
- **NID lookup:** hardcoded — NID `5102284394` returns an existing client (Name=Mahfuzul Islam, CIF=553577, Email=lmahfuzul@gmail.com). All other NIDs treated as new client.
- **Application submission:** persisted to MySQL or PostgreSQL; tracking number returned per the format `IDLC-YYMMDD-XXXX`
- **Eligibility check:** server-side mirror of the client-side calculation in the prototype (interest rate 13.00% p.a., 70%-of-income load threshold, BDT 25 lakh minimum reduced amount)

Languages: English + বাংলা. All ~250 translation strings are provided in `translations.js`. No additional translation work is required.

2.2 Out of scope — Phase 1

- Real SMS gateway integration → **Phase 2**
 - Real CIF / NID lookup against [Customer]'s core banking system → **Phase 2**
 - Document upload, OCR, e-signature, branch verification → **Phase 3**
 - Customer Login flow (currently inert in the prototype) → **Phase 3**
 - Push notifications, email automation → **Phase 3**
 - Mobile native apps → **out of scope entirely**
 - Branding refresh (logo, colours, content) — handled by [Customer]'s in-house team
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3. Sitemap & User Flows

3.1 Sitemap

```
index.html (marketing home)
├─ floating "Apply for Loan"
│   └─ apply.html (Loan for Individuals | Loan for Businesses)
│       ├── apply-individual.html
│       │   └─ modal: Purpose of Loan × Financing Mode →
│       │       ├── apply-home-loan.html
│       │       ├── apply-car-loan.html
│       │       └─ apply-personal-loan.html
│       └─ apply-business.html
│           ├── modal: Desired Product × Financing Mode → apply-sme-loan.html
│           └─ external link → https://idlc.com/corporate-loan (new tab)
```

3.2 Critical user flow — individual loan application

1. Land on home → click floating "Apply for Loan" (bottom-right, sticky)
2. See 2 options (Individual / Business) → click Individual
3. See 3 sub-options → click (e.g.) Home Loan
4. Pop-up modal: pick Purpose + Financing Mode → click Proceed
5. Land on product page → enter Name, NID, DOB (Day/Month/Year dropdowns), Phone → click Request OTP
6. 4 OTP boxes appear → enter 1234 (stub) → click Submit OTP
7. Page scrolls to top of application-form card
8. If NID matches existing-client stub → Name, CIF, Email auto-fill silently
9. Fill remaining fields + answer "Loan Burden?" Yes/No (conditional Monthly Loan Burden) + tick 2 declarations → click Submit Application
10. Eligibility check runs (see §4.1):
 - eligible → green success + tracking number
 - reduced (eligible amount ≥ BDT 25 lakh) → green success + tracking number; message uses reduced wording
 - ineligible → amber warning, no tracking number issued

3.3 SME flow

Identical shape to §3.2 but launched from `apply-business.html` → SME card → modal dropdowns are Product (3 options) + Mode (2 options) → land on `apply-sme-loan.html`.

3.4 Corporate flow

Click "Loan for Corporates" on `apply-business.html` → browser opens `https://idlc.com/corporate-loan` in a new tab. No internal page.

4. Business Rules

4.1 Eligibility calculation

Parameter	Value
Standard interest rate (eligibility math)	13.00% per annum
EMI formula	$P \times r \times (1+r)^n / ((1+r)^n - 1)$ where $r = 13/12/100$ and $n = \text{years} \times 12$
Eligibility threshold	Total monthly load $\leq$ 70% of declared monthly income
Total monthly load	Projected EMI + existing monthly loan burden (if Yes selected)
Reduced-amount minimum	BDT 25,00,000 (25 lakh) — below this, application is rejected
Currency	BDT throughout

Decision matrix:

Condition	Status	Confirmation message wording	Tracking issued?
Total load at requested amount $\leq$ 70% of income	eligible	"...the requested loan amount of BDT [requested]..."	Yes
Total load at requested amount $>$ 70%, but max-eligible at same tenor $\geq$ BDT 25 lakh	reduced	"... a loan amount of BDT [reduced]..."	Yes
Existing burden alone $>$ 70% of income, OR max-eligible $<$ BDT 25 lakh	ineligible	"Your monthly income will not cover your loan burden and IDLC's risk appetite..."	No

4.2 OTP

- Phase 1: hardcoded 1234
- Phase 2: integrate with a Bangladesh-licensed SMS gateway (vendor to recommend; common options: SSLWireless, SMS Express, Robi Axiata corporate SMS)
- OTP expiry: 5 minutes
- Max retry attempts: 3 per phone per hour
- Rate limit on OTP requests: max 5 per phone per hour

4.3 Existing-client (NID) lookup

- Phase 1: hardcoded mapping — only NID `5102284394` returns an existing client
- Phase 2: integrate with [Customer]'s core banking / CIB API
- UX behaviour identical in either phase: silent auto-fill of Name + CIF + Email on the application form

4.4 Other rules

- Minimum applicant age: **18 years at the date of OTP request**
- NID pattern: 10, 13, or 17 digits (text input, JavaScript-validated)
- Phone pattern: `01XXXXXXXX` (11 digits, Bangladesh mobile)
- Both declaration checkboxes must be ticked before Submit Application can fire
- Tracking number format: `IDLC-YYMMDD-XXXX` (where XXXX is 4-char alphanumeric, uppercase, excluding 0, O, 1, I, L to avoid visual ambiguity)
- Tracking number generated only for `eligible` and `reduced` outcomes

5. API Contract

All endpoints return JSON, use bearer-token authentication post-OTP, and emit HTTP 400 (`{field, message}` array) on validation errors and HTTP 5xx (`{error_id}`) on server errors.

Endpoint	Method	Purpose	Phase 1 behaviour
<code>/api/otp/request</code>	POST	Initiate OTP for a phone	Returns <code>{success: true, expires_in: 300}</code> . No SMS dispatched.
<code>/api/otp/verify</code>	POST	Verify entered OTP	Compares against <code>1234</code> . Returns <code>{success: true, token}</code> or <code>{success: false}</code> .
<code>/api/customer/lookup</code>	GET	NID-based existing-client check	Hardcoded mapping. Returns <code>{exists: true, name, cif, email}</code> for <code>5102284394</code> , else <code>{exists: false}</code> .
<code>/api/applications</code>	POST	Submit full application	Persists to DB, runs eligibility, returns <code>{tracking_number, status, eligible_amount}</code> .
<code>/api/applications/:tracking</code>	GET	Status lookup (Phase 1: read-only, Phase 3: full status)	Returns current row.

Request and response schemas to be detailed by vendor in their technical design document, due 1 week after kickoff.

6. Data Model

```
applications
  id (PK, UUID)
  tracking_number (UNIQUE, VARCHAR 20)
  product (ENUM: home-loan, car-loan, personal-loan, sme-loan)
  purpose (VARCHAR 120)
  financing_mode (ENUM: Conventional, Islamic)
  name, nid (indexed), dob (DATE), phone (indexed)
  cif (NULLABLE – only set for existing-client matches)
  email, profession, monthly_income (DECIMAL 15,2)
  has_burden (BOOL), monthly_burden (DECIMAL, NULLABLE)
  expected_amount (DECIMAL), loan_term_years (INT)
  eligibility_status (ENUM: eligible, reduced, ineligible)
  eligible_amount (DECIMAL)
  language (ENUM: en, bn)
  declarations_accepted_at (TIMESTAMP)
  ip_address (VARCHAR 45)
  user_agent (TEXT)
  created_at, updated_at (TIMESTAMP)

audit_log
  id, application_id (FK), event_type, actor, timestamp, payload (JSON)

otp_requests
  id, phone, code_hash, requested_at, expires_at, verified_at (NULLABLE)

admin_users
  id, email (UNIQUE), password_hash, role, created_at, last_login_at
```

Vendor to propose final DDL in technical design document.

7. Acceptance Criteria

Each AC is demonstrable in under 60 seconds in the staging environment.

#	Criterion
AC-1	Open / → disclaimer ribbon visible at top; brand, lang toggle, hero, stat strip, news cards, why-us band, quick links, footer all render; floating Customer Login + Apply for Loan pinned bottom-right; scrolling does not move them.
AC-2	Click বাংলা on any page → all visible UI translates → click any internal link → still Bengali → refresh → still Bengali.
AC-3	From /apply-individual.html click Home Loan → modal opens with house sketch, blank Purpose dropdown (6 options), blank Financing Mode dropdown (2 options). Pick "Apartment Purchase" + "Conventional Financing" → click Proceed → land on apply-home-loan.html?purpose=Apartment+Purchase&mode=Conventional . Title reads "Home Loan Conventional Financing". Chip below reads "Apartment Purchase".
AC-4	From apply-business.html click Loan for SME → modal opens with storefront sketch, dropdowns Purnota / Working Capital / Long Term Loan × Conventional / Islamic Financing.
AC-5	From apply-business.html click Loan for Corporates → new browser tab opens https://idlc.com/corporate-loan .
AC-6	On any product page, enter NID 5102284394 , DOB valid date ≥18 yrs, valid BD phone → Request OTP → 4 OTP boxes appear → enter 1234 → Submit OTP → loan form reveals with Name="Mahfuzul Islam", CIF="553577", Email="lmahfuzul@gmail.com" pre-populated.
AC-7	Same flow with any NID other than 5102284394 → after OTP, loan form reveals with Name = whatever was typed in OTP gate, CIF and Email blank.
AC-8	Pick DOB making applicant <18 → click Request OTP → red toast: "Applicants must be at least 18 years of age to submit this loan application." OTP does not fire.
AC-9	DOB Day=31, Month=February, Year=2000 → Day auto-clamps to 29 (leap year). Change Year to 2001 → Day clamps to 28.
AC-10	After OTP, enter Income=200,000; Has Burden=No; Expected Amount=20,00,000; Loan Term=5 → tick both declarations → Submit. Result: green success, message reads "... requested loan amount of BDT 20,00,000...", tracking number visible.
AC-11	After OTP, enter Income=1,00,000; Has Burden=No; Expected Amount=5,00,00,000; Loan Term=5 → Submit. Result: green success, message reads "... a loan amount of BDT [computed reduced]...", tracking number visible.
AC-12	After OTP, enter Income=30,000; Has Burden=Yes; Monthly Loan Burden=25,000; Expected Amount=25,00,000; Loan Term=5 → Submit. Result: amber warning, message reads "Your monthly income will not cover your loan burden and IDLC's risk appetite...", no tracking number.
AC-13	Try to Submit Application without ticking both declarations → red toast: "Please accept both declarations to continue." Focus moves to first unticked checkbox.
AC-14	Type 2500000 into any BDT field → formats live to 25,00,000 . Mid-string editing preserves cursor position.

#	Criterion
AC-15	Type 100000000 (10 crore) into EMI calculator Loan Amount → result rows update within 100ms; no upper cap.
AC-16	Admin login → see paginated applications list, search by tracking number, view row detail. Export current page as CSV with all columns.
AC-17	Submit any eligible application → confirmation rendered in current language → toggle to বাংলা → all confirmation text (including tracking-number block) re-renders in Bengali.

8. Commercial

8.1 Phasing & quotation

Vendor is asked to quote **Phase 1 (fixed price)** at this stage. Phase 2 and 3 will be scoped and quoted separately after Phase 1 acceptance.

8.2 Payment milestones (suggested 30/40/30)

Milestone	Trigger	%
M1 — Kickoff	Signed contract + technical design document submitted	30%
M2 — Phase 1 acceptance	All 17 ACs demonstrated on staging environment	40%
M3 — Warranty close	30 days post-acceptance, no Sev-1 / Sev-2 bugs outstanding	30%

8.3 Technical stack

Vendor must pick from the following — no exotic frameworks without prior approval:

- **Backend:** PHP/Laravel **or** Node.js/Express **or** Python/Django
- **Frontend:** keep the existing static HTML/CSS/JavaScript stack from the prototype. No SPA framework (no React, Vue, Angular) required.
- **Database:** MySQL 8+ **or** PostgreSQL 14+
- **Hosting:** Linux VPS (DigitalOcean droplet, AWS Lightsail, or equivalent). Sub-\$20/month tier acceptable for Phase 1 staging.
- **No proprietary CMS** (no Magento, Adobe Experience Manager, Oracle, SAP, etc.) without [Customer] approval.

8.4 Source-code ownership

All source code, design assets, database schemas, deployment scripts, and account credentials produced under this engagement are work-for-hire and become **[Customer]'s exclusive property on final**

payment for the corresponding phase. The vendor retains no licence, no right to reuse the code or designs, and no right to host any artefact beyond the warranty period.

8.5 Warranty

30 days from Phase 1 acceptance. Vendor fixes any Sev-1 (application non-functional), Sev-2 (major flow broken), and Sev-3 (cosmetic / minor) defect at no additional charge. Sev-1 acknowledgment within 4 working hours.

8.6 Project management

[Customer] expects:

- Single point of contact (project lead) at vendor
- Weekly status email (max half-page)
- Staging URL accessible to [Customer] from end of week 2 onwards
- No requirement for Gantt charts, Jira dashboards, or formal PM artefacts

End of document.